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$$C \xrightarrow{\text{fum. } HNO_3} CO_2$$

$$2H \xrightarrow{\text{fum. } HNO_3} H_2O$$

$$S \longrightarrow SO_2 \xrightarrow{O + H_2O} H_2SO_4$$

The resulting solution is treated with excess of $BaCl_2$ solution H_2SO_4 present in the solution in quantitatively converted into $BaSO_4$, from the mass of $BaSO_4$, the mass of sulphur and hence the percentage of sulphur in the compound can be calculated.

Procedure:

A known mass of the organic compound is taken in clean carius tube and added a few mL of fuming HNO_3. The tube is the sealed. It is then placed in an iron tube and heated for about 5 hours. The tube is allowed to cool to temperature and a small hole is made to allow gases produced inside to escape. The carius tube is broken and the content collected in a beaker. Excess of $BaCl_2$ is added to the beaker H_2SO_4 acid formed as a result of the reaction is converted to $BaSO_4$. The precipitate of $BaSO_4$ is filtered, washed, dried and weighed. From the mass of $BaSO_4$, percentage of S is found.

Mass of the organic compound = w g

Mass of the $BaSO_4$ formed = x g

233g of $BaSO_4$ contains 32 g of Sulphur

$\therefore$ x g of $BaSO_4$ contain $\left(\frac{32}{233} \times \frac{x}{w}\right)$ g of S

Percentage of Sulphur = $\left(\frac{32}{233} \times \frac{x}{w} \times 100\right)\%$

Example -2

In an estimation of sulphur by carius method, 0.2175 g of the substance gave 0.5825 g of $BaSO_4$ calculate the percentage composition of S in the compound.

Weight of organic compound 0.2175 g

Weight of $BaSO_4$ 0.5825 g

233 g of $BaSO_4$ contains 32 g of S

0.5825 g of $BaSO_4$ contains $\frac{32}{233} \times \frac{0.5825}{0.2175}$

Percentage of S=$\frac{32}{233} \times \frac{0.5825}{0.2175} \times 100$

= 36.78 %

Evaluate Yourself

6) 0.16 g of an organic compound was heated in a carius tube and H_2SO_4 acid formed was precipitated with $BaCl_2$. The mass of $BaSO_4$ was 0.35g. Find the percentage of sulphur [30.04]

Estimation of halogens: carius method:

A known mass of the organic compound is heated with fuming HNO_3 and $AgNO_3$. C,H &S get oxidized to CO_2, H_2O & SO_2 and halogen combines with $AgNO_3$ to form a precipitate of silver halide.

$$X \xrightarrow[AgNO_3]{\text{fum.}HNO_3} AgX\downarrow.$$

The ppt of AgX is filtered,

washed, dried and weighed. From the mass of AgX and the mass of the organic compound taken, percentage of halogens are calculated.